

Luke Cullen

772-341-2067 | lukeculleninc@gmail.com | lucullen.dev | linkedin.com/in/luke-cullen-319701305 | github.com/retromonos

EDUCATION

University of Central Florida

Bachelor of Science in Computer Science - Minor in Data Science - 3.72 GPA

Orlando, FL

August 2023 – May 2027

EXPERIENCE

Web Developer Intern

October 2025 – Present

CDL Techrangers

Orlando, FL

- Recently accepted an offer for the **CDL Techrangers**, an organization which develops **educational software** that impacts **10,000+ UCF students annually**.
- Expected to **onboard and maintain** pre-existing educational solutions alongside a team of **27 developers and fellow interns**, including **Flask** and **PhP** based infrastructure.
- Taking initiative to **brainstorm, plan and implement** new web apps for University students and faculty.
- Developing **methods** and **marketing pitches** to **increase adoption** of our products across and beyond the University.

Hackathon Organzier

July 2025 – Present

Knight Hacks

Orlando, FL

- Helped utilize over **\$5,000** worth of event funding through planning meals, ensuring dietary restriction compliance to boost **accessibility**.
- Authored **sub-event infrastructure** and accompanying management interface, including statistic tracking, **boosting engagement** for **1000+** people attending the event.
- Implemented a Discord bot to consistently remind hackers about ongoing events, **prioritizing quick readability**.
- Coordinated various activities during the event, such as socials and our career fair layout, **improving crowd control** and **maximizing participation**.

Lead Developer, Vice President

May 2025 – Present

DenTraceX

Orlando, FL

- Co-founded a non-profit organization** dedicated to collecting dental records of missing persons, in order to cross-reference them with unidentified remains.
- Authored a **full-stack React application** from scratch, using **ExpressJS**, **PostgreSQL**, as well as our own self-hosted **Minio S3 bucket** storage solution.
- Constructed a **novel** code-based representation for jaw/tooth records, allowing easy searching/filtering to identify **real-world** human remains, intended for use by medical examiners and independent investigators.
- Working with a team of diverse disciplines to process and handle **sensitive medical information** and **maintain legal compliance**.

PROJECTS

VL-ADK | *Google ADK, Python, NVIDIA Nemotron, YOLO-E, React, FastAPI*

September 2025

- Created at ShellHacks 2025, won **1st place** for **NVIDIA's Hack the Future Challenge**, **2nd place** for **Microsoft's AI for Good Challenge**, and **3rd place** for **Waymo's Reimagining Navigation Challenge**.
- Mimicks the functionality of a **generalized VLA robotics model** using a fleet of **Google ADK** agents, powered by **NVIDIA Nemotron**.
- Allows **dynamic prompting** of a cost-effective **NVIDIA Jetson Orin**-powered robot, achieving **intelligent movement and spatial reasoning**, even with extremely limited sensing capabilities.
- With the assistance of a **YOLO-E** model for object detection, accomplished **structured paths** and **object search/navigation** behaviors, completely operated by **autonomous AI agents**.

Voiceboard | *React, TypeScript, Gemini API, Mermaid*

September 2024

- Created at ShellHacks 2024, won **1st place** for **Google's Hack for Social Good Challenge**.
- Authored the whiteboarding interface, allowing the entire app to be voice-controlled through **speech recognition** and **Google's Gemini AI** to massively improve accessibility.
- Implements prompt engineering through the **Gemini API** to create **context aware Mermaid diagram code**.

Lootcode | *React, Clerk, TypeScript, MySQL, Zx Shell, Docker, tRPC*

February 2024 – May 2024

- Created for **Knight Hacks Project Launch 2024**, won **1st place**.
- Developed a **gamified study application** for Data Structures and Algorithms, featuring **100+ unique problems**.
- Live-generates **Docker containers** to self-host code grading instances, maintaining **server security**.
- Accumulated **50,000+** page visits and **500+** monthly active users, featured in **Linux Magazine's** OSS section.

TECHNICAL SKILLS

Languages: Java, Python, C/C++/C#, Lua, SQL, TypeScript, HTML/CSS

Libraries/Frameworks: React, ExpressJS, FastAPI, S3, Prisma, Clerk, JWT, tRPC, MySQL, PostgreSQL, Node.js, Tailwind CSS, Websockets, OpenCV, MediaPipe, pandas, YOLO-E, Google Gemini, NVIDIA Nemotron, Google ADK

Developer Tools: Linux, Coolify, Git, Docker, VS Code, Eclipse, Cursor AI